

Research Article

Foundation or Formula? A Simulation-Based Comparison of Google's TimesFM-3 and Classical Time-Series Models

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Abstract

Time-series foundation models such as Google's TimesFM-3 forecast series they were never trained on. Whether they should replace classical methods such as ARIMA and exponential smoothing is difficult to settle on public benchmarks, because few benchmark datasets are absent from every model's pre-training corpus. This exploratory study describes, under controlled conditions, how TimesFM-3 behaves relative to classical methods and in which regimes it may be preferred. The evaluation data are generated from nine known processes, so the model cannot have seen the series. Across 7 200 series and a pre-registered protocol with a 12-step horizon, neither family dominates. TimesFM-3 is never more than 1.31 times worse than the best method in a cell, whereas every classical method is at least 2.2 times worse somewhere; the largest classical failures occur with only two seasonal cycles, where the automatic methods fall back to non-seasonal models, and without those cells automatic ARIMA's worst case is 1.62. With four or more cycles automatic ARIMA is 21–24% more accurate than TimesFM-3 on seasonal ARIMA data. Among five foundation models, this robustness is specific to TimesFM-3, and it is specific to short horizons: at 48 steps, TimesFM-3 and the other foundation models forecast a decline on a saturating process whose level stays flat, and automatic ARIMA has the smaller worst case elsewhere. On intermittent demand TimesFM-3 shows no advantage over simple benchmarks built from the history. In post-hoc checks, randomised parameters, heavy-tailed noise and outliers leave these patterns qualitatively unchanged, and on the official test period of 1 000 M4 monthly series TimesFM-3 is level with the fifth- to seventh-ranked M4 entries, behind the four best, and among ten methods whose ranks cannot be separated. Because TimesFM-3 is a black box, the study characterises its behaviour, not the reasons for it; the findings are conditional on the processes and horizons examined, and the model's weights are licensed for non-commercial use only.

1 | Introduction

Time-series foundation models are large neural networks, pre-trained on many series and then applied to new series without task-specific training. On 31 August 2026 Google Research released TimesFM-3, a 330-million-parameter decoder-only transformer and the first member of its family designed for multivariate forecasting, which its developers report as the top-ranked pre-trained model on three public benchmarks [1], a claim open to the leakage concerns discussed next.

For applied forecasting the relevant question is whether such a model should be used, in preference to a classical method such as ARIMA, for the series at hand; the two families also differ in cost, interpretability, the basis of their prediction intervals and the terms under which they may be used (Section 8).

Published benchmarks are poorly suited to answering that question. Of 401 datasets used to evaluate 22 time-series foundation models, only about 6% had never appeared in any model's pre-training corpus, and even a 0.1% overlap can move the mean absolute percentage error by 8 to 29 percentage points [2]. Foundation-model evaluations have also been criticised for comparing against under-tuned classical baselines rather than the combinations that win forecasting competitions [3]. This study therefore generates its own evaluation data. Under a known data-generating process (DGP) the foundation model cannot have seen the series, because they did not exist before the experiment, and the model families searched by AutoARIMA and AutoETS contain the true process of the linear and exponential-smoothing DGPs. What the design excludes is contamination at the level of the realisation; what it cannot exclude is familiarity at the level of the family, because TimesFM-3 is pre-trained partly on synthetic series [1]. Such familiarity cannot be measured here, and its direction cannot be assumed: synthetic corpora contain autoregressive and seasonal structure, resembling D1–D5, but also level shifts, piecewise trends and sparse spikes, resembling D6–D8. No conclusion of this paper relies on its direction. The aim of the study is exploratory: to describe, under controlled conditions, how the accuracy of TimesFM-3 relative to classical methods varies across regimes, and to derive diagnostic baselines for when it may be preferred and when it should not. It proposes no new estimator, test or theory. The design can establish what TimesFM-3 does but not why: the 330 million parameters have no individual interpretation and the pre-training corpus cannot be inspected, so explanations of its behaviour offered here are interpretations consistent with the measurements, not findings. The measurements concern one model, one checkpoint and univariate zero-shot use, and they do not support general statements about foundation models as a class.

TimesFM-3 is compared with seasonal naive, Theta, AutoETS, AutoARIMA and an equal-weight combination of the last three on 7 200 series from nine DGPs, four lengths from 24 to 200 observations and three horizons, under a protocol written and frozen before any forecast was produced; every departure from it is recorded. This pre-registered comparison is confirmatory in design; everything after it is post hoc and exploratory, and is labelled as such. The evaluation is then extended with stronger benchmarks and two further foundation models, tested for representativeness and robustness, and repeated on 1 000 M4 monthly series. Section 2 describes TimesFM-3, Section 3 the design and Section 4 the pre-registered results. Section 5 extends the evaluation, Section 6 examines representativeness and robustness, and Section 7 turns to M4. Section 8 covers operational considerations, Section 9 discusses the evidence, and Section 10 concludes; reproducibility details are in Section 11.

1.1 | Related work

TimesFM [4], Chronos [5], Moirai [6], TimeGPT [7] and Lag-Llama [8] established that a single pre-trained network can forecast unseen series zero-shot, and GIFT-Eval [9] and fev-bench [10] supplied common benchmarks that include AutoARIMA, AutoETS and seasonal naive among their baselines. The critical literature has grown alongside: Meyer, Kaltenpoth, Zalipski, and Müller [2] on contamination, and Adler, Chang, Draxler, Abdi, and Smyth [11], who found on six public datasets that the calibration error of five foundation models grows with forecast length. The present study differs from the latter in using generated rather than public data, in testing accuracy differences with multiplicity control rather than only summarising calibration, and in evaluating TimesFM-3. Applied comparisons on real data are appearing in several fields—Chu, Szeto, and Chung [12] compare six foundation models with SARIMA on influenza surveillance data—but they cannot say why a model works, because the generating mechanism is unknown.

The design follows a pattern established in this journal, a controlled simulation over several DGPs followed by an empirical application [13, 14, 15], and it checks the realism of the simulated series in a feature space, as in Kang, Hyndman, and Smith-Miles [16] and Talagala, Hyndman, and Athanasopoulos [17]. AutoARIMA and AutoETS follow the Hyndman–Khandakar procedures [18, 19], Theta [20] won M3 [21], and the combination is included because combinations are the standard against which competition entries are judged [21, 3]. Evaluation follows the scale-free conventions of Hyndman and Koehler [22] and the M5 competition [23, 24] and avoids the pitfalls catalogued by Hewamalage, Ackermann, and Bergmeir [25].

2 | The TimesFM-3 model

This section describes the architecture and pre-training data of TimesFM-3 (Section 2.1) and the checkpoint and configuration used (Section 2.2).

2.1 | Architecture and pre-training data

TimesFM-3 is the third generation of the decoder-only TimesFM family [4, 1]. Each input series is normalised by an iterative reversible instance normalisation, with linear detrending of strongly trending inputs, and divided into non-overlapping patches of 32 time points, each embedded as a token; 20 transformer layers with model dimension 1280 and 16 attention heads give about 330 million parameters (the checkpoint loaded here has 330 710 976). Layers of causal attention along time alternate with layers of attention across variates, so target series and covariates can be processed jointly, and a contiguous patch masking scheme produces the whole horizon in one forward pass, in output patches of 64 points. For each step the model returns nine quantiles, the 10th to the 90th percentile, and its point forecast is the median [1, 26, 27]. These details come from the developers’ blog post and model card and from the configuration file of the checkpoint (accessed 23 September 2026); none has yet been described in a peer-reviewed publication. The pre-training corpus of more than one trillion time points combines the GIFT-Eval pre-training collection, Wikipedia page views to November 2023, Google Trends queries to the end of 2022, and synthetic and augmented series [26].

2.2 | Checkpoint and configuration

The weights (google/timesfm-3.0-pytorch on the Hugging Face Hub, revision 43046b8) are loaded with version 3.0.1 of the timesfm Python package, which provides both the timesfm and timesfm3 modules [27, 26], and are licensed for non-commercial use only (Section 8). The model is used zero-shot: it receives only the raw context, with no fine-tuning, covariates or seasonal period, through predict_batch with its default settings (no symmetric averaging, no positivity constraint, 32-bit arithmetic). Its point forecast is the model’s own median output; the nine deciles are then sorted, which matters only where they cross (Section 5.3). A context shorter than the model’s working length is left-padded with zeros and masked, so at $n = 24$ the model sees one partly filled patch; the package’s maximum context of 15 360 points truncates no series in this study, and a series forecast alone or inside a batch of other lengths gives quantiles that agree to within 2×10^{-5} . Because no quantile outside the 10th and 90th percentiles is available, the widest interval that can be formed is the nominal 80% interval. The weights, code and revision are public, so every forecast can be regenerated (Section 11); the developers’ own evaluator settings are examined in Section 5.3.

3 | Design of the simulation study

This section describes the data-generating processes (Section 3.1), the forecasting methods compared (Section 3.2), and the evaluation measures and inferential procedures (Section 3.3). Table 1 summarises the design in the aims–data–estimands–methods–performance form recommended for simulation studies by Morris, White, and Crowther [28].

TABLE 1 | Structure of the simulation study.

Element	Specification
Aims	Describe how the accuracy of zero-shot TimesFM-3 relative to classical methods varies across known data-generating regimes; exploratory.
Data-generating mechanisms	Main design: nine processes (Table 2) with fixed parameters, $n \in \{24, 48, 96, 200\}$, $H = 12, 200$ replications per cell. Robustness design (Section 6.2): parameters drawn per series from wide ranges (Table S8), four versions (Gaussian, heavy-tailed, outliers, tested seasonality), 200 replications per cell.
Estimands	For each method and cell: the expected MASE over $h = 1, \dots, 12$ (and RMSSE, scaled pinball loss, 80% coverage); for each pair of methods and cell: the pseudo-median of the paired MASE difference. Summaries across cells (worst ratio, win counts) were defined post hoc.
Methods	Seasonal naive, Theta, AutoETS, AutoARIMA, their combination and TimesFM-3 (pre-registered); DOTM, the M4 Comb benchmark, a second combination, TimesFM-2.5, Chronos-Bolt and the Croston family (post hoc).
Performance measures	Mean MASE with its Monte Carlo standard error; ratio to the best method in a cell, with bootstrap intervals for summaries across cells; paired Wilcoxon tests with Hodges–Lehmann estimates and Benjamini–Hochberg control; coverage and width of 60% and 80% intervals. With 200 replications the Monte Carlo standard error of a paired MASE difference is about 2% of the mean (Section 4.2).

Note: The mean MASE (descriptive) and the pseudo-median difference (inferential) are different estimands and can disagree when the error distribution is skewed, as on the intermittent process D8 (Section 4.2); both are reported.

3.1 | Data-generating processes

Nine processes were chosen to span the range from “a classical model is exactly right” to “no classical model is right”. Table 2 lists them. Each series is generated at length $n + H$ with $H = 12$; the first n observations form the context supplied to every method, and the final 12 are held out as the truth. One realisation of each process is shown in Figure 1.

TABLE 2 | The nine data-generating processes and their parameters.

ID	Data-generating process	Parameters	Correctly specified by
D1	AR(1)	$\phi = 0.7$	ARIMA
D2	ARMA(1,1)	$\phi = 0.6, \theta = 0.4$	ARIMA
D3	ARIMA(1,1,1) with drift	$\phi = 0.5, \theta = 0.3, \text{drift } 0.3$	ARIMA
D4	SARIMA(1,0,0)(1,1,0) ₁₂	$\phi = 0.6, \Phi = 0.5$	ARIMA
D5	ETS(A,A,A)	$\alpha = 0.3, \beta = 0.01, \gamma = 0.2$	ETS
D6	Local level with a break	break at $0.7n$, jump 10σ	neither
D7	Logistic growth	$K = 180, r = 0.08, \text{floor } 20$	neither
D8	Intermittent demand	$p = 0.3, \text{size } 1 + \text{Pois}(4)$	neither
D9	AR(1) with GARCH(1,1) errors	$\phi = 0.6, a = 0.1, b = 0.85$	ARIMA in mean

Note: The last column states which classical family is correctly specified by construction; for D6–D8 none is, and for D9 the conditional mean is correctly specified while the conditional variance is not. Innovations are Gaussian with standard deviation $\sigma = 2$ ($\sigma = 3$ in D7; D8 has none). D1, D2 and D9 fluctuate about a mean of 100 after a burn-in of 300 steps (D9: $\omega = 0.1$); D3 starts at 100 after its differences are burnt in; D4 starts from the cycle $100 + 10 \sin(2\pi k/12)$ plus noise, its seasonal differences burnt in; D5 starts from level 100, trend 0.2 and seasonal states $10 \sin(2\pi k/12)$; in D6 the level is a random walk from 100 with innovation standard deviation 0.5 and σ is that of the observation noise; D7 has lower asymptote 20; in D8 a demand occurs with probability p .

The parameter values in Table 2 were fixed in the pre-registered protocol, each chosen so that its process represents its regime unambiguously. The linear processes D1–D4 and the conditional mean of D9 use coefficients no larger than 0.7 in absolute value, well inside the stationarity and invertibility regions [29]. The smoothing parameters of D5 lie inside the usual region $0 < \beta < \alpha < 1, 0 < \gamma < 1 - \alpha$ [30], with a small trend parameter so that the level stays positive. The break in D6 equals ten noise standard deviations, so the comparison concerns adaptation to the break rather than its detection [31]; D7 follows the logistic curve, the standard model of saturating growth [32]. In D8 the mean interval between demands is $1/0.3 \approx 3.3$ periods and the squared coefficient of variation of the sizes is $4/25 = 0.16$, so against the cut-offs of Syntetos, Boylan, and Croston [33] the process is intermittent rather than lumpy, the regime for which Croston’s method was developed [34]. The GARCH(1,1) errors of D9 [35] have persistence $a + b = 0.95$, strongly persistent but covariance-stationary [36], a standard benchmark specification in volatility forecasting [37]. Every series has a level of about 100, or a positive lower asymptote in D7, so percentage errors are defined except where zeros are intrinsic (D8); the trend parameter of D5 and the asymptote of D7 were adjusted for this reason before any forecast was produced. Because the conclusions could depend on these values, Section 6 repeats the comparison with parameters drawn at random. Three design choices need a word. The break in D6 is placed inside the context, at $0.7n$: a break in the held-out window is unforecastable by any method, whereas one in the context asks whether a method forecasts from the new level or reverts to the old one. The inflection of D7 sits at $0.6(n + H)$, so the held-out window lies in the saturating arm, where a method that fits a linear drift overshoots. And the shortest length, $n = 24$, leaves a seasonal model only two cycles from which to estimate its seasonal structure, which separates whether a model is correctly *specified* from whether it is *estimable*. Two hundred replications per (DGP, length) cell give $9 \times 4 \times 200 = 7\,200$ series, each seeded from its own identifiers.

3.2 | Forecasting methods

No method is tuned by hand; each classical model uses its own automatic selection, the comparison a practitioner faces. The classical models are fitted with StatsForecast [38] at its default settings, stated because the results depend on them. AutoARIMA searches stepwise over $p, q \leq 5$ and $P, Q \leq 2$ with $p + q + P + Q \leq 5$, chooses $d \leq 2$ and $D \leq 1$ by unit-root tests, allows a mean or drift and selects by AICc; AutoETS searches the full error–trend–season space (ZZZ) with damped and undamped trends by AICc; Theta is the standard theta method with classical multiplicative seasonal adjustment [39]. The equal-weight combination of Theta, AutoETS and AutoARIMA is included because the classical state of the art is a combination [3, 40]. Its point forecast is the mean of the members’ forecasts and its quantiles the decile-by-decile mean of theirs (vincentization), which gives a narrower distribution than a mixture when the members disagree.

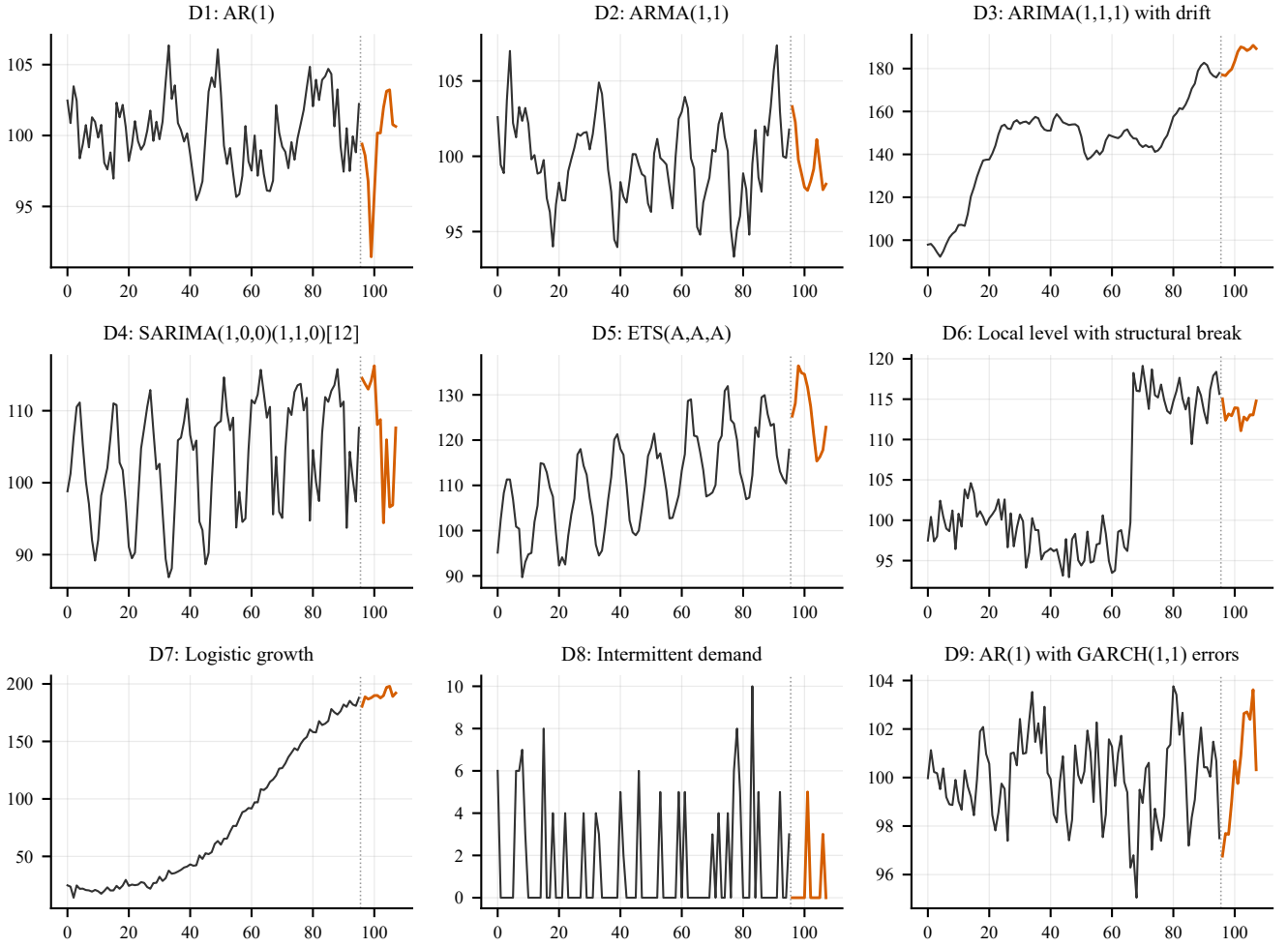


FIGURE 1 | One realisation of each data-generating process at $n = 96$. The context supplied to every method is in grey and the held-out horizon in colour, separated by the dotted line.

Every classical method receives the true seasonal period ($m = 12$ on D4 and D5, $m = 1$ elsewhere), whereas TimesFM-3 receives only the raw context. This reflects ordinary practice—an analyst usually knows that data are monthly—but it favours the classical methods on D4 and D5 once enough cycles are observed; with two cycles they rarely use it (Section 4.3). Section 6 examines the alternative of testing for seasonality.

3.3 | Evaluation measures and statistical inference

Point accuracy is measured mainly by the mean absolute scaled error [MASE; 22]. With context $y_1, \dots, y_T$ ($T = n$), horizon H and point forecasts $\hat{y}_{T+h}$,

$$\text{MASE} = \frac{\frac{1}{H} \sum_{h=1}^H |y_{T+h} - \hat{y}_{T+h}|}{\frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|}, \quad (1)$$

so the scale is the in-sample seasonal-naive error for the seasonal processes and the naive error for the others ($m = 12$ throughout for the M4 tier, as in M4). The root mean squared scaled error (RMSSE) replaces absolute by squared errors in both numerator and denominator and takes the square root, and sMAPE is the symmetric percentage error of the M4 competition. MAPE is undefined when an actual value is zero, which happens only on D8 (in 9.96% of all evaluations over the three horizon slices), and is reported only in the Supporting Information (Section S1) [22, 41]. Probabilistic accuracy is measured by a strictly proper scoring rule [42], the pinball loss over the nine deciles $\tau \in \{0.1, \dots, 0.9\}$ scaled by the same

denominator,

$$\text{SPL} = \frac{\frac{1}{9H} \sum_{\tau} \sum_{h=1}^H \max\{\tau(y_{T+h} - \hat{q}_{T+h}^{(\tau)}), (1 - \tau)(\hat{q}_{T+h}^{(\tau)} - y_{T+h})\}}{\frac{1}{T-m} \sum_{t=m+1}^T |y_t - y_{t-m}|}, \quad (2)$$

where $\hat{q}_{T+h}^{(\tau)}$ is the τ -quantile forecast, together with the coverage and width of the nominal 60% and 80% intervals [43]. Wider intervals cannot be formed from TimesFM-3's nine deciles, so $[q_{0.1}, q_{0.9}]$ is the widest interval compared; classical quantiles are taken from intervals at levels 20, 40, 60 and 80, which map onto the same deciles. Within each (DGP, length) cell and horizon slice ($h = 1, 1-6$ and $1-12$), TimesFM-3 is compared with each classical method by a paired Wilcoxon signed-rank test [44] on per-replication MASE differences, with the Hodges–Lehmann estimate [45] and the median MASE ratio as effect sizes, and the Benjamini–Hochberg procedure at a false discovery rate of 0.05 within each opponent family of 108 tests (36 cells and three slices), as pre-registered [46]; the results report the $h = 1-12$ slice. The Diebold–Mariano test [47] is not used: the replications are independent, so there is no serial dependence for it to model, and in the real-data tier three rolling origins per series leave it without power even with small-sample corrections [48]. The pre-registered protocol had reserved it for that tier; the departure is recorded.

4 | Simulation results

This section reports the pre-registered comparison: overall accuracy and the pairwise tests (Section 4.1), effect sizes (Section 4.2), the role of estimability (Section 4.3), the seasonal and intermittent regimes (Section 4.4), and interval coverage and the combination rule (Section 4.5). Statements about TimesFM-3 describe the behaviour of its forecasts; where an explanation is offered, it is an interpretation. Figure 2 shows TimesFM-3 relative to the best classical method in each of the 36 (process, length) cells; the mean MASE of every method in every cell, over $h = 1, \dots, 12$ and 200 replications, is given in Table S9 of the Supporting Information.

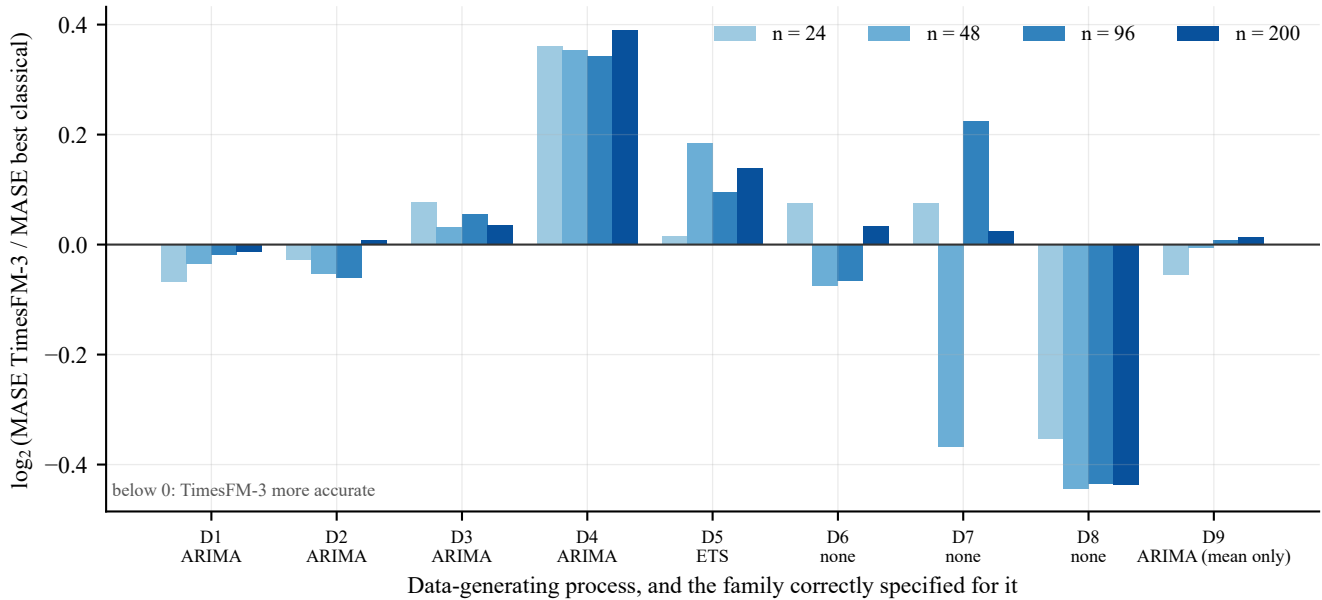


FIGURE 2 | Accuracy of TimesFM-3 relative to the best classical method in each cell, on a $\log_2$ scale. Bars below zero mark cells where TimesFM-3 is more accurate. The label under each process names the family correctly specified for it by construction.

4.1 | Overall accuracy and pairwise comparisons

TimesFM-3 has the lowest mean MASE in 16 of the 36 (DGP, length) cells and the classical methods in 20: AutoARIMA 10, seasonal naive 4, and two each for Theta, AutoETS and the combination. Against the best classical method of each cell, 18 differences survive Benjamini–Hochberg correction, 7 favouring TimesFM-3 and 11 the classical method; because that

comparator is selected on the same replications, these p -values are descriptive only. The inferential claims rest on the 180 comparisons with a fixed opponent (Table 3): 132 are significant, 113 favouring TimesFM-3. These counts summarise the tests; they weight every cell equally and depend on the mix of processes, and pooling the tests into a single family changes them by at most two (Supporting Information, Section S6). **AutoARIMA is the only classical method that TimesFM-3 does not clearly beat**; against every other method, the combination included, TimesFM-3 wins the large majority of significant comparisons.

TABLE 3 | Significant pairwise comparisons of TimesFM-3 with each classical method at $h = 1, \dots, 12$.

Opponent	Significant comparisons	TimesFM-3 wins	Opponent wins
AutoARIMA	20	11	9
Seasonal naive	26	22	4
AutoETS	27	25	2
Theta	29	28	1
Combination	30	27	3

Note: Paired Wilcoxon signed-rank tests in the 36 (process, length) cells at $h = 1, \dots, 12$, Benjamini–Hochberg corrected within each opponent family of 108 tests (three horizon slices).

Figure 3 shows the effect behind each count: against AutoARIMA most intervals straddle 1, and the cells where AutoARIMA is ahead are the seasonal processes from $n = 48$, whereas against the other methods most intervals lie wholly on the side of TimesFM-3.

4.2 | Effect sizes and sensitivity to the choice of test

The median Hodges–Lehmann shifts in MASE across the 36 cells are -0.010 against AutoARIMA, -0.091 against the combination, -0.133 against seasonal naive, -0.183 against AutoETS and -0.227 against Theta (negative favours TimesFM-3), so against AutoARIMA the typical difference is about one hundredth of a MASE unit. The Monte Carlo standard error of the paired difference between TimesFM-3 and AutoARIMA is 0.8% to 4.8% of AutoARIMA’s mean MASE (median 2.1%), so differences smaller than about 6% are unlikely to be detected in a cell: the 11–9 split against AutoARIMA sits at the resolution of the design and does not show that the two methods are equivalent. A paired t -test agrees with the Wilcoxon test in 163 of 180 comparisons (90.6%). The largest block of disagreements is on the intermittent process D8 against seasonal naive: the t -test finds TimesFM-3 better at every length ($p < 0.0001$), the signed-rank test finds no difference ($p = 0.11$ to 0.92). The mean difference is driven by a minority of series on which seasonal naive fails badly, whereas on 64.5–72.0% of series TimesFM-3 is marginally less accurate—the two tests estimate different quantities of a heavily skewed error distribution (Table 1).

4.3 | Correct specification versus estimability

Pre-registered hypothesis H1 held that correctly specified classical models would win on their home ground, most clearly at short lengths; **it is not supported in that form**. On D1 and D2 TimesFM-3 is at least as accurate as the correctly specified AutoARIMA at every length except D2 at $n = 200$, the only significant difference. Correct specification bought AutoARIMA nothing detectable on short and moderate series. A post-hoc *Oracle ARIMA*, which knows the true orders and estimates only the parameters by Gaussian maximum likelihood, is consistent with order selection being the cost: on D1–D3 at $n = 24$ AutoARIMA pays a 9.9% to 13.9% penalty for selecting the orders, while the oracle beats TimesFM-3 in 15 of the 16 ARIMA cells (for example 1.351 against 1.417 on D1 at $n = 24$). On D4 at $n = 24$ most oracle fits did not converge (156 of 200, used as returned), so that cell is not interpreted.

The seasonal processes show the effect most clearly. With two complete cycles ($n = 24$), AutoETS scores a mean MASE of **3.03 on D5, data an ETS process generated**, against 1.14 for seasonal naive and 1.15 for TimesFM-3: the correctly specified model has 2.7 times the error of the naive rule. The failure is not an over-fitted seasonal model but the absence of one: given $m = 12$ and 24 observations, AutoETS chose a seasonal form for 8.5% of the D4 series and none of the D5 series, and AutoARIMA for none (Supporting Information, Table S11), whereas seasonal naive applies the known period without estimating anything. The operative distinction is therefore whether a model is *estimable* from the series at hand, not whether it is correctly specified. Table 4 quantifies the resulting asymmetry with each method’s mean MASE

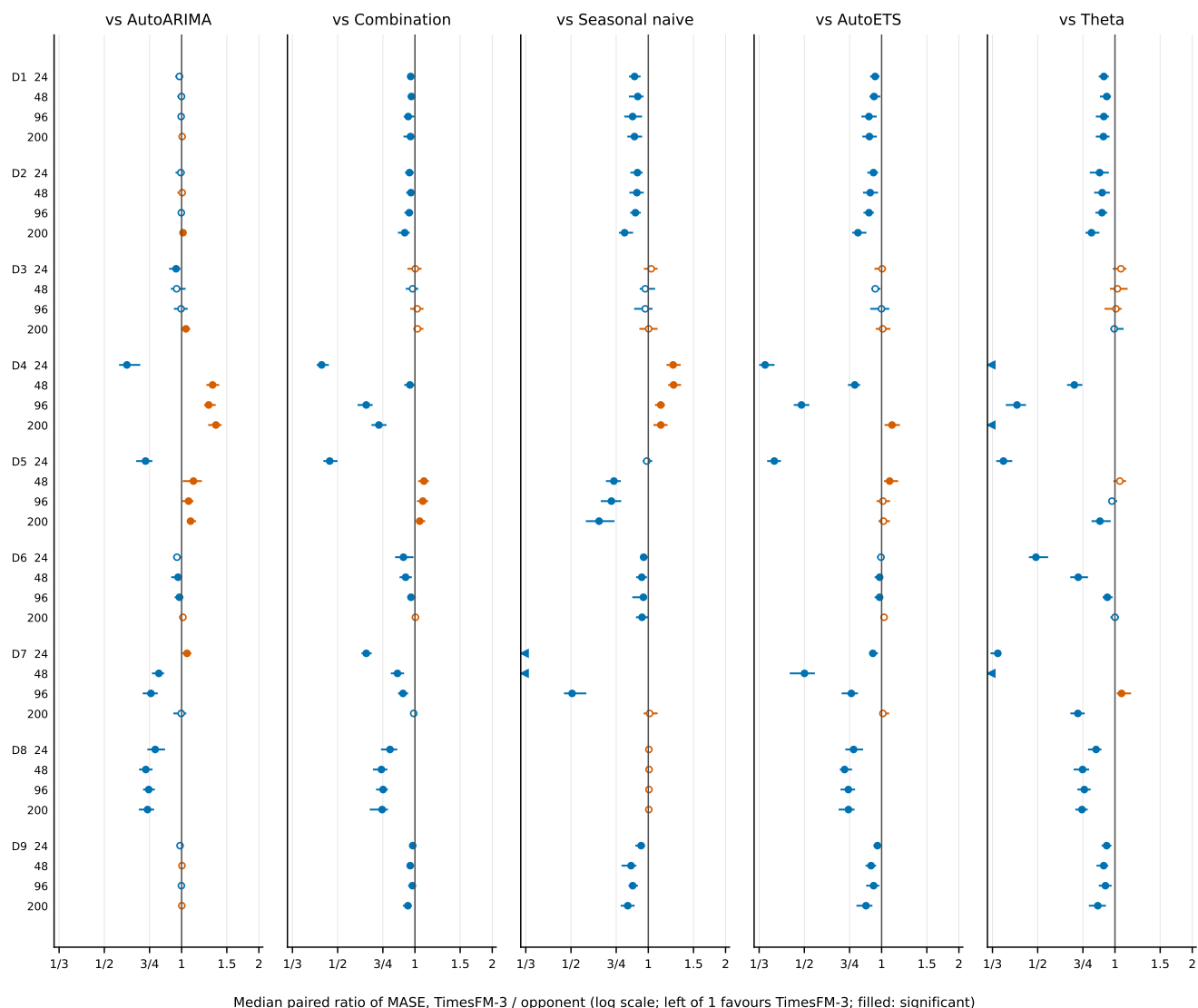


FIGURE 3 | Pairwise comparisons of TimesFM-3 with each classical method, cell by cell: median of the paired per-series ratios of MASE over $h = 1, \dots, 12$, with 95% bootstrap intervals, on a logarithmic scale. Filled markers are significant after Benjamini-Hochberg correction; points left of 1 favour TimesFM-3. Arrowheads mark medians beyond the plotted range.

as a multiple of the best in the same cell—a relative measure, because D3 is hard for every method and would dominate absolute worst cases. **TimesFM-3 is never worse than 1.31× the best method of a cell**, and within 0.8% of it in the median cell, whereas every classical method is at least 2.2× worse somewhere: AutoARIMA 2.22× and AutoETS 3.50× (both on D4 at $n = 24$), Theta 4.69×, seasonal naive 5.65×. Leaving out the two cells with two seasonal cycles, TimesFM-3’s worst ratio is unchanged and AutoARIMA’s falls to 1.62 (on D7 at $n = 96$), AutoETS’s to 2.72 and the combination’s to 2.08; the last two come from AutoETS on D4 at $n = 96$, a cell in which StatsForecast’s estimation fails (Section 4.4), and with the value of R’s forecast package there AutoETS’s worst ratio would be 2.13, on D7 at $n = 48$. The worst ratio is a descriptive summary defined after the protocol, and it depends on the methods compared, notably on seasonal naive with the known period (Section 6.2).

4.4 | Seasonal and intermittent regimes

Strong seasonality with sufficient data (D4). At $n = 48, 96$ and 200 AutoARIMA’s mean MASE is 21.8%, 21.1% and 23.7% below TimesFM-3’s (adjusted $p < 0.0001$), and D5 shows the same more mildly (4% to 12%). Once at least four cycles are observed, AutoARIMA recovers the seasonal structure and TimesFM-3 does not match it; seasonal naive, which estimates nothing, is also more accurate than TimesFM-3 on D4 at every length (1.005 to 1.053 against 1.124 to 1.321).

TABLE 4 | Mean MASE relative to the best method in the same cell, summarised over the 36 cells.

Method	Median	Mean	All 36 cells		Without the two-cycle cells	
			Worst	Cell	Worst	Cell
Seasonal naive	1.200	1.470	5.65	D7 $n = 48$	5.65	D7 $n = 48$
Theta	1.251	1.614	4.69	D4 $n = 200$	4.69	D4 $n = 200$
AutoETS	1.194	1.366	3.50	D4 $n = 24$	2.72	D4 $n = 96$
AutoARIMA	1.032	1.137	2.22	D4 $n = 24$	1.62	D7 $n = 96$
Combination	1.108	1.273	3.00	D4 $n = 24$	2.08	D4 $n = 96$
TimesFM-3	1.008	1.054	1.31	D4 $n = 200$	1.31	D4 $n = 200$

Note: Each entry divides a method's mean MASE by the lowest mean MASE in the same (process, length) cell, so a value of 1.00 means that the method was the best in that cell; the mean is arithmetic. The two-cycle cells are D4 and D5 at $n = 24$. Geometric means and 90th percentiles are in the Supporting Information, Table S10.

With two cycles the ranking among the others reverses: TimesFM-3 beats AutoARIMA by 42% (1.321 against 2.284). Theta deteriorates on D4 as the series lengthens (Theta 1.94 at $n = 48$ and 4.16 at $n = 200$). D4 is seasonally integrated, so its seasonal pattern drifts, whereas classical decomposition estimates one fixed pattern from the whole history; with additive instead of multiplicative decomposition the failure is almost unchanged (in a check on 60 replications, Theta's mean MASE at $n = 200$ was 3.83 against 3.95), so it concerns the model form rather than the adjustment. Because the classical results could depend on the implementation, the D4 and D5 cells were refitted with `auto.arima`, `ets` and `thetaf` of the R package `forecast` [18], the reference implementation of the Hyndman–Khandakar procedures (post hoc; Supporting Information, Table S13). R reproduces every failure discussed here—AutoARIMA 2.87 and AutoETS 3.95 on D4 at $n = 24$, AutoETS 3.30 on D5 at $n = 24$, Theta 4.09 on D4 at $n = 200$ —and, like StatsForecast, almost never fits a seasonal model with two cycles. Where both choose seasonal forms the two implementations agree closely, with one exception: on D4 at $n = 96$ StatsForecast's AutoETS has a mean MASE of 2.41 against 1.04 for R, because its optimiser settles on a near-zero seasonal smoothing parameter where R finds a high one. That cell is an implementation failure rather than a property of the model.

Intermittent demand (D8). On MASE TimesFM-3 reduces error against the best general-purpose method by 21.7% to 26.5% at the four lengths (adjusted $p < 0.0001$) and has the lowest pinball loss of the pre-registered methods, but it *loses* on RMSSE at every length (Supporting Information, Table S6); sMAPE, which scores a zero forecast of a zero outcome as perfect, is not informative here. About 70% of D8's observations are zero: MASE is minimised by the conditional median, which is zero or near it, RMSSE by the conditional mean, which is positive, so a median-like forecast wins MASE and loses RMSSE [41, 49]. The same holds against six purpose-built intermittent-demand methods, added post hoc: Croston's classic and optimised variants, the Syntetos–Boylan approximation [50], ADIDA [51], IMAPA [52] and TSB [53], the last with StatsForecast's fixed constants $\alpha_d = \alpha_p = 0.2$ (Table S7). TimesFM-3 beats all six on MASE at every length (24/24 significant; median ratios 0.70 to 0.82), while the Croston family has the lower RMSSE (0.694–0.769, against 0.774–0.806 for TimesFM-3; AutoARIMA is lowest at $n = 200$). Section 5.4 shows that both of TimesFM-3's advantages on D8 are matched by simple benchmarks built from the history.

Across horizons ($h = 1, 1\text{--}6$ and $1\text{--}12$) every method degrades and no regime-average ranking reverses (Supporting Information, Figure S1). Averaged within regimes, TimesFM-3 has the lowest mean MASE even among the correctly specified processes, because a few catastrophic classical cells (Theta at 4.16 on D4 at $n = 200$, AutoETS at 3.03 on D5 at $n = 24$) move a mean far more than several narrow classical wins.

4.5 | Coverage of intervals and the combination rule

Averaged over the processes (Figure 4), TimesFM-3's 80% coverage stays between 0.802 and 0.834 at every length, on average across lengths closer to nominal than any other method, and it does not deteriorate with the horizon (0.786 at $h = 1$, 0.814 over $h = 1\text{--}12$). Averaging lets over- and under-coverage cancel, however; the mean per-cell absolute deviation from nominal is 0.047 for TimesFM-3, 0.066 for AutoARIMA, 0.089 for the combination, 0.101 for AutoETS, 0.127 for Theta and 0.144 for seasonal naive, and TimesFM-3's per-cell 80% coverage ranges from 0.583 to 0.921. No method delivers reliable 80% intervals on all nine processes. The classical intervals rest on Gaussian innovations, violated by construction on D8 (Bernoulli occurrence with Poisson sizes) and D9 (conditionally Gaussian but unconditionally heavy-tailed errors). An interval score, which rewards narrow intervals and penalises outcomes outside them (80% interval, scaled as MSIS), puts TimesFM-3 first in 22 of the 36 cells (Supporting Information, Table S12). On the structural break D6

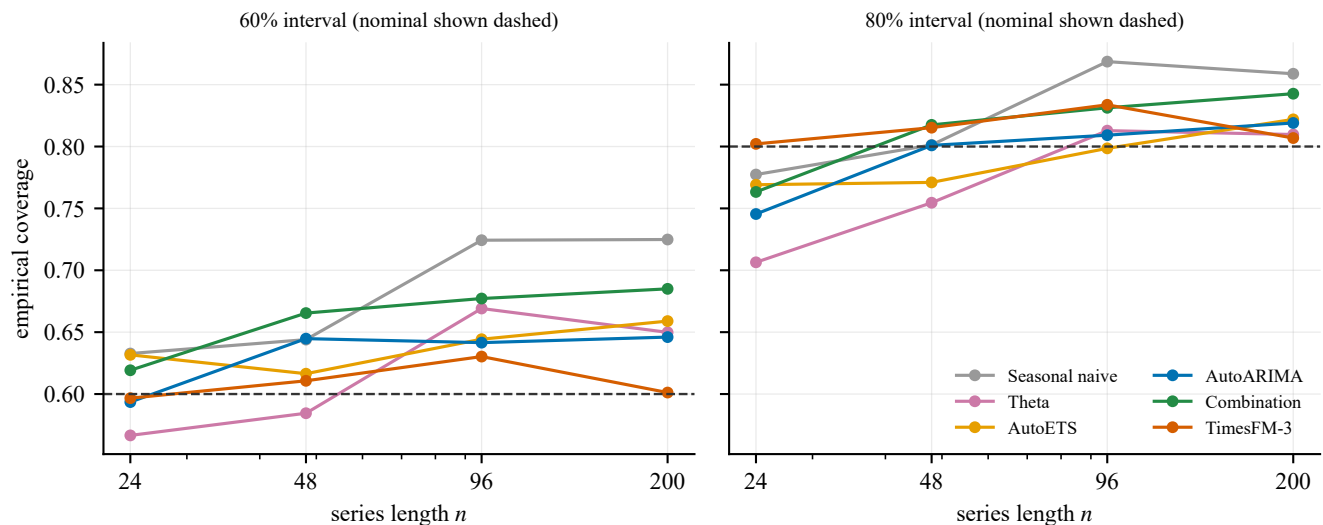


FIGURE 4 | Empirical coverage of the nominal 60% and 80% intervals, averaged over all nine processes, against series length. The nominal level is dashed. TimesFM-3's 80% coverage stays within 0.802–0.834 across every length; the classical methods under-cover at $n = 24$ and over-cover from $n = 96$.

every classical method covers 0.966 to 0.999, because the break inflates the estimated variance and the intervals become very wide (scaled widths 4.8 to 10.4); TimesFM-3 covers 0.828 to 0.889 with intervals about half as wide (3.0 to 5.1) and has the lowest interval score at every length.

TimesFM-3 beats the pre-registered mean combination in 27 of 30 significant comparisons. Because the mean is not robust to a failing member such as Theta, a *median* combination of the same three methods was also tried (post hoc), in the spirit of Petropoulos and Svetunkov [54]; it is slightly worse (mean MASE 1.712 against 1.656, TimesFM-3 1.439), and TimesFM-3 still wins 25 of 29 significant comparisons. Under absolute error the mean combination can never be worse than the average of its members, by convexity, a guarantee the median does not carry.

5 | Extended evaluation

This section asks how precise the summaries of Section 4 are (Section 5.1), whether stronger benchmarks and other foundation models change them (Section 5.2), how much they depend on the configuration of TimesFM-3 (Section 5.3), and what the intermittent-demand result means once simple benchmarks are included (Section 5.4), and what changes at a longer horizon (Section 5.5). All of it is post hoc.

5.1 | Monte Carlo uncertainty

The Monte Carlo standard error of a cell's mean MASE is 1.5% to 6.2% of the mean (median 3.2%; Supporting Information, Table S2). Bootstrapping the replications within each cell (2 000 draws, paired across methods), the worst ratio of TimesFM-3 to the cell-best method is 1.31 (95% interval 1.28 to 1.36) and that of AutoARIMA, the smallest among the classical methods, 2.22 (2.07 to 2.38); their difference has an interval of 0.75 to 1.05. Because the best method of a cell is chosen on the same replications, the statistic could be biased upwards; choosing it on one half of the replications and estimating the ratio on the other gives 1.32, so the bias is small. These intervals reflect Monte Carlo error only, not the choice of processes and parameters, which Section 6 varies. The count of cells won is less stable—TimesFM-3's 16 has an interval of 13 to 20—and no conclusion rests on it.

5.2 | Stronger classical benchmarks and other foundation models

Six methods were added (Table 5): dynamic optimised Theta [DOTM; 55]; the Comb benchmark of M4, the mean of simple, Holt and damped exponential smoothing on seasonally adjusted data [3]; a combination of AutoETS, AutoARIMA and DOTM; TimesFM-2.5 [56] and Chronos-Bolt (Base, 205 million parameters) [5, 57], both zero-shot and Apache-2.0

licensed; and TimesFM-3 with its developers' evaluator settings (Section 5.3). TimesFM-2.5 was run with the same evaluator settings (symmetric averaging, positivity, quantile-crossing correction) and a context limit of 512 points, which truncates no simulated series, so its matched comparison is with the evaluator-settings row of TimesFM-3; Chronos-Bolt was run with its defaults. Three further configurations were added afterwards: Chronos-2 [58] and TiRex [59], two newer zero-shot models with 120 and 35 million parameters, both run with their package defaults, and TimesFM-2.5 with flip invariance and positivity switched off, which matches the default settings of the main TimesFM-3 runs.

TABLE 5 | Robustness of fifteen methods across the 36 cells of the main design.

Method	Worst ratio to best	Mean ratio to best	Within 10% (%)	Cells won
TimesFM-3, evaluator settings	1.27 [1.26, 1.33]	1.058 [1.055, 1.066]	81	5 [2, 8]
TimesFM-3	1.31 [1.29, 1.36]	1.063 [1.060, 1.072]	78	0 [0, 4]
TimesFM-2.5	2.18 [2.04, 2.33]	1.121 [1.116, 1.130]	67	7 [2, 8]
TiRex	2.24 [2.06, 2.45]	1.121 [1.116, 1.131]	64	0 [0, 4]
TimesFM-2.5, settings off	2.26 [2.10, 2.44]	1.124 [1.120, 1.134]	67	3 [0, 5]
AutoARIMA	2.22 [2.08, 2.39]	1.131 [1.125, 1.142]	64	8 [5, 10]
Chronos-2	2.75 [2.56, 2.97]	1.139 [1.135, 1.150]	67	4 [3, 6]
Combination (ETS, ARIMA, DOTM)	2.99 [2.84, 3.16]	1.241 [1.236, 1.253]	47	1 [0, 2]
Combination (Theta, ETS, ARIMA)	3.00 [2.85, 3.17]	1.243 [1.237, 1.255]	44	1 [1, 3]
Chronos-Bolt	3.06 [2.95, 3.25]	1.296 [1.291, 1.309]	58	1 [0, 2]
AutoETS	3.50 [3.28, 3.73]	1.312 [1.304, 1.327]	31	1 [0, 2]
Seasonal naive	5.70 [5.47, 6.03]	1.347 [1.337, 1.362]	25	2 [1, 3]
DOTM	4.66 [4.34, 4.99]	1.413 [1.405, 1.429]	22	1 [1, 2]
M4 Comb	4.45 [4.18, 4.79]	1.456 [1.447, 1.474]	19	1 [0, 2]
Theta	4.69 [4.37, 5.03]	1.476 [1.467, 1.493]	19	1 [1, 2]

Note: Main design, 7 200 series, mean MASE over $h = 1, \dots, 12$, ratios to the best of the fifteen methods in each of the 36 (process, length) cells. Brackets: 95% intervals from 500 bootstrap resamples of the replications within each cell. Mean ratio is the geometric mean over cells. Among the original six methods alone, the worst ratio of TimesFM-3 is 1.31 [1.28, 1.36] and that of AutoARIMA 2.22 [2.05, 2.38].

The stronger classical benchmarks do not change the picture. DOTM and the Comb benchmark share Theta's failure on D4, where a classical decomposition with fixed seasonal indices meets a seasonal pattern that evolves, and replacing Theta by DOTM leaves the combination's worst ratio at 2.99. TimesFM-3 wins 28 of 29 significant comparisons against DOTM and all 29 against the Comb benchmark; AutoARIMA remains the only classical method it does not clearly beat (11 wins to 9). **Among the five foundation models, the robustness belongs to TimesFM-3 alone** (Table 5): TimesFM-2.5 has a worst ratio of 2.18 (on D7 at $n = 24$), and 2.26 with its settings switched off, so the difference is not one of configuration; TiRex has 2.24, Chronos-2 2.75 (on D6 at $n = 24$) and Chronos-Bolt 3.06, against 2.22 for AutoARIMA. In the median cell every foundation model but Chronos-Bolt is within 2% to 4% of the best, and TimesFM-2.5 is the best method in 7 cells, more than any model but AutoARIMA (8). Like TimesFM-3, the newer models do not clearly beat AutoARIMA (Chronos-2 10–10, TiRex 9–12). Friedman tests with Nemenyi critical differences [60] within each regime agree with the pairwise tests: TimesFM-3 is tied for the best mean rank on the linear processes (with TimesFM-2.5, TiRex and AutoARIMA) and on the break and saturation processes (with no other method), and AutoARIMA has the best mean rank on the seasonal processes, tied only with TimesFM-3 under its evaluator settings. Averaged over the design, the 80% intervals of TimesFM-3 cover 0.814 (Monte Carlo SE 0.003), TimesFM-2.5 0.812, Chronos-2 0.789, TiRex 0.842, Chronos-Bolt 0.834 and AutoARIMA 0.794 (Supporting Information, Table S3, by process for the models of the first comparison).

5.3 | Configuration and output of TimesFM-3

The developers' benchmark evaluator averages the forecasts of a series and its negation and constrains non-negative series to non-negative forecasts, whereas the main study used the package defaults. With the evaluator settings the worst ratio falls from 1.31 to 1.27 and the geometric mean ratio over the twelve methods from 1.059 to 1.054; no conclusion changes. The output is a 12×9 array of deciles whose raw median equals the point forecast exactly; quantile crossing occurs in 32% to 46% of steps on D8 and essentially never elsewhere, so only on D8 does it matter that the main runs take the raw median as point forecast and the evaluator-settings runs the median after sorting. Given the month of the year as past-and-future covariates ($\sin 2\pi t/12$, $\cos 2\pi t/12$) on D4 and D5—the information the classical methods receive through their period—TimesFM-3's mean MASE rises in seven of the eight cells (for example 1.124 to 1.194 on D4 at $n = 96$), so supplying the

period in this form did not close AutoARIMA's seasonal advantage. Using the mean of the deciles instead of the median as point forecast changes no process by more than 0.01 except D6 (0.835 to 0.875) and D8.

5.4 | Intermittent demand re-examined

Because MASE rewards the conditional median, which on D8 is near zero, the trivial all-zero forecast belongs in the comparison, and because D8 is independent over time, so do benchmarks that simply use the empirical distribution of the history [61]: its mean as point forecast and its deciles as predictive distribution. Table 6 adds them, with the scaled mean error (bias) and periods in stock [PIS; 62], which accumulates forecast errors as a stock position would.

TABLE 6 | Intermittent demand (D8) re-examined with simple benchmarks, bias and stock measures.

Method	MASE	RMSSE	sME	sPIS	SPL
All-zero forecast	0.668–0.718	0.785–0.825	−0.72 to −0.67	−91 to −81	–
History: mean; empirical deciles	0.924–0.975	0.692–0.744	−0.07 to −0.02	−13 to −3	0.282–0.314
TimesFM-3	0.683–0.780	0.774–0.806	−0.67 to −0.57	−81 to −74	0.288–0.325
TimesFM-3, mean of deciles	0.915–1.002	0.694–0.760	−0.06 to −0.01	−10 to −4	0.288–0.325
TimesFM-2.5	0.678–0.770	0.777–0.806	−0.67 to −0.59	−81 to −74	0.289–0.324
Chronos-Bolt	0.708–0.786	0.758–0.789	−0.59 to −0.54	−72 to −65	0.294–0.328
Chronos-2	0.681–0.750	0.776–0.808	−0.66 to −0.63	−81 to −76	0.290–0.323
TiRex	0.683–0.776	0.771–0.800	−0.67 to −0.58	−81 to −73	0.289–0.324
Croston-SBA	0.921–1.025	0.694–0.763	−0.04 to 0.05	−7 to 2	–
ADIDA	0.926–0.972	0.700–0.753	−0.09 to −0.02	−16 to −5	–
TSB	0.916–0.971	0.716–0.764	−0.10 to 0.00	−17 to −1	–
AutoARIMA	0.924–0.996	0.698–0.765	−0.05 to −0.03	−12 to −3	0.341–0.380

Note: Process D8, 200 replications per length; ranges over the four lengths $n \in \{24, 48, 96, 200\}$. sME: mean error (forecast minus actual) scaled by the MASE denominator; negative values mean under-forecasting. sPIS: periods in stock [62], cumulated over the horizon and scaled by the mean demand of the context; large negative values mean persistent stock-outs. SPL: scaled pinball loss over the nine deciles, defined only for methods with a predictive distribution. TimesFM-3, mean of deciles, uses the average of its nine deciles as point forecast. History: the mean of the context as point forecast and the empirical deciles of the context as predictive distribution; the median of the context is zero or near it, so its point forecast would repeat the all-zero row.

The all-zero forecast has the lowest MASE at every length (0.668 to 0.718), below TimesFM-3's median forecast (0.683 to 0.780). The MASE advantage of Section 4.4 is thus the advantage of a forecast close to zero: its bias (−0.67 to −0.57 MASE units) and PIS (−81 to −74) are close to those of the all-zero forecast and imply persistent stock-outs, so it is not evidence of useful accuracy. With the mean of its deciles as point forecast, TimesFM-3 has an RMSSE of 0.694 to 0.760 and a bias within 0.06 of zero, level with the best Croston-family method and with the mean of the history (0.692 to 0.744). Its pinball loss, 0.288 to 0.325, is about 15% below that of the classical methods (0.341 to 0.380), but their quantiles come from Gaussian intervals that put probability on negative demand; the empirical deciles of the history score 0.282 to 0.314, slightly lower than TimesFM-3 at every length. **On intermittent demand, therefore, TimesFM-3 shows no advantage over simple benchmarks:** it behaves much like the empirical distribution of the history, and its advantage over the classical methods reflects their Gaussian intervals. The other foundation models, Chronos-2 and TiRex included, behave the same way (Table 6). The result concerns a process that is independent over time, the case most favourable to history-based benchmarks.

5.5 | A longer horizon

Calibration of foundation models is reported to degrade with the horizon [11], and the main design stops at 12 steps. The nine processes were therefore regenerated with the same seeds for a 48-step horizon and forecast by the six pre-registered methods, TimesFM-2.5 and Chronos-2 (Supporting Information, Section S7 and Figure S3). Because the inflection of D7 sits at $0.6(n + H)$, the longer horizon moves it into the context for $n = 200$, so that the series has already flattened when the forecast starts. Over the first 12 steps the picture of Section 4 is recovered: without D7, TimesFM-3's worst ratio to the best method is 1.31 and AutoARIMA's 2.11. Further out it is not. Over steps 25 to 48, **AutoARIMA has the smaller worst case** (1.41 against 1.80, D7 excluded), although TimesFM-3 remains closer to the best method in the median cell (1.075 against 1.109). On D7 at $n = 200$ every foundation model forecasts a decline while the process stays at its plateau near 200: TimesFM-3's mean forecast falls to 128 by step 48, TimesFM-2.5's to 98 and Chronos-2's to 146, and their mean

MASE over the 48 steps is 8.37, 12.22 and 5.30, against 1.11 for seasonal naive and 2.51 for AutoARIMA. What in the models produces the decline cannot be established here. The coverage of TimesFM-3's 80% intervals falls from 0.788 over steps 1 to 12 to 0.760 over steps 25 to 48, and AutoARIMA's from 0.760 to 0.720, so both drift away from nominal, as Adler, Chang, Draxler, Abdi, and Smyth [11] report for foundation models. The robustness of Section 4.3 is thus a property of short horizons.

6 | Representativeness and robustness of the design

Simulation conclusions are conditional on the processes and parameters used. This section asks how closely the simulated series resemble real ones (Section 6.1) and whether the conclusions survive random parameters and departures from the assumptions (Section 6.2). Both analyses are post hoc.

6.1 | Representativeness of the simulated series

Following Kang, Hyndman, and Smith-Miles [16], each series is summarised by four features: normalised spectral entropy (forecastability), STL trend strength, STL seasonal strength (period 12) and the first autocorrelation of the first differences. The simulated series at $n = 96$ and the 1 000 M4 Monthly series of Section 7, each cut to the last 96 observations of its training period, are placed in this space with the features standardised on the M4 series. An M4 series is *covered* when some simulated series lies at least as close to it as the 95th percentile of the distance from an M4 series to its nearest other M4 series.

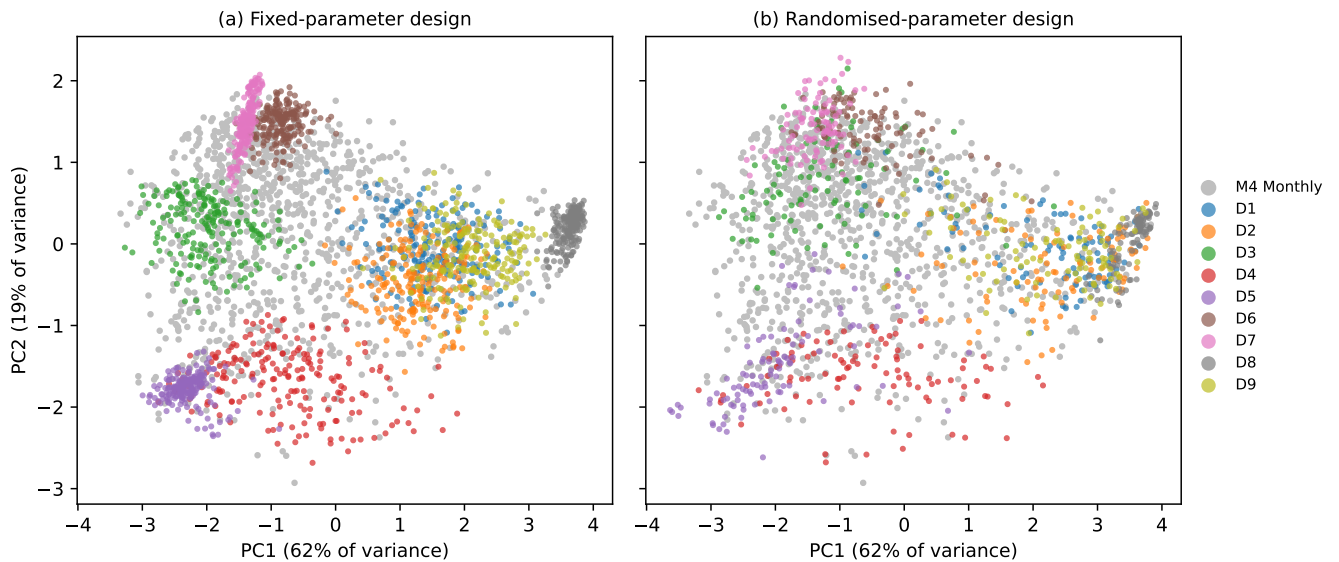


FIGURE 5 | Simulated and M4 Monthly series in a four-feature instance space (spectral entropy, trend strength, seasonal strength and first-order autocorrelation of the differences), first two principal components. Grey points are M4 series; coloured points are simulated series at $n = 96$. (a) The fixed-parameter design covers 48.5% of the M4 series; (b) the randomised-parameter design covers 75.8%, with coverage defined in Section 6.1.

The fixed design covers 48.5% of the M4 series (bootstrap interval 45.2% to 51.4%; Figure 5): each process forms a compact cluster and the regions between them, which hold many real series, are empty. The randomised design of Section 6.2 covers 75.8% (73.2% to 78.4%). Across thresholds from the 80th to the 99th percentile, lengths of 48, 96 and 200, and a six-feature space adding the first autocorrelation and lumpiness, the randomised design covers 76% to 89% of M4 at the 95th percentile and never less than the fixed design, also when the latter is subsampled to equal size; conversely, 60% to 92% of the randomised series lie within the same distance of some M4 series, fewest at $n = 200$. The parameter ranges were set before this statistic was first computed. M4 series are more strongly trending than the simulated ones (median trend strength 0.87 against 0.63), and only 1.5% lie nearest to D8, because M4 has no intermittent series; the intermittent-demand results rest on the simulation alone.

6.2 | Randomised parameters and departures from the assumptions

Every series now draws its own parameters uniformly from wide ranges containing the fixed values of Table 2 (Supporting Information, Table S8): autoregressive coefficients up to 0.95, intermittency from lumpy ($p = 0.1$) to nearly continuous ($p = 0.6$), breaks of three to fifteen noise standard deviations in either direction. Four versions use the same 7 200 parameter draws (200 per process and length): *random parameters* with Gaussian innovations; *heavy tails*, Student- t_3 innovations of the same variance built from the same normal draws (negative binomial sizes on D8); *outliers*, each context point shifted with probability 0.03 by four to six robust standard deviations (a demand multiplied by five on D8), with the held-out values left clean; and *tested seasonality*, in which the classical methods use $m = 12$ only if the 90% autocorrelation test of the M4 benchmarks [3] finds seasonality, applied when at least three cycles are observed, and $m = 1$ otherwise. TimesFM-3 never receives a period, so its forecasts are unchanged in the last version.

TABLE 7 | Headline conclusions under randomised parameters and departures from the assumptions.

	Random	Heavy tails	Outliers	Tested seas.
TimesFM-3 cell wins (of 36)	14	15	16	16
TimesFM-3 median ratio to cell-best	1.029	1.038	1.004	1.017
TimesFM-3 worst ratio to cell-best	1.56	1.54	1.49	1.55
Smallest worst ratio, classical	2.12	2.32	2.09	1.58
Significant comparisons, won–lost	93–29	100–28	101–27	100–27
D4, $n \geq 48$: AutoARIMA better (of 3)	3	3	2	3
D8: Croston better on RMSSE (of 4)	4	4	4	4
D5, $n = 24$: AutoETS / seasonal naive ($m = 12$)	2.03	2.05	1.91	2.25

Note: 200 replications per (process, length) cell, 7 200 series per column; every series draws its own parameters from the ranges in Table S8 of the Supporting Information. Ratios use mean MASE over $h = 1, \dots, 12$. Significant comparisons are paired Wilcoxon tests of TimesFM-3 against each of the five general-purpose classical methods in each cell (180 per column), Benjamini–Hochberg at 0.05 within each (column, opponent) family of 36 cells (the main design’s families also include the three horizon slices). The D8 row compares TimesFM-3 with the best of Croston-SBA, TSB and ADIDA; the last row divides by seasonal naive with the true period in every column. In the tested-seasonality column the classical methods use $m = 12$ only where the M4 seasonality test finds seasonality; TimesFM-3’s forecasts are those of the first column.

In the three versions with a known period the conclusions were qualitatively consistent with the main design (Table 7), although TimesFM-3’s worst ratio lies outside the main design’s bootstrap interval. TimesFM-3 is the best method in 14 to 16 cells and within 4% of the best in the median cell; its worst ratio rises to 1.49 to 1.56 (bootstrap intervals up to about 1.75), always on the logistic process D7 at $n = 96$, while the smallest classical worst ratio is 2.09 to 2.32. With the Benjamini–Hochberg family per opponent, TimesFM-3 wins 93 to 101 of 180 comparisons and loses 27 to 29, AutoARIMA again being the opponent it does not clearly beat (7–7, 6–7, 10–4 and 8–6 across the four versions). AutoARIMA’s advantage on D4 with at least four cycles persists (median over the three lengths 21% to 26%; with outliers it is ahead at two of the three lengths, by 15% and 5%), and the Croston family keeps the lower RMSSE on D8 in every version.

When the classical methods test for seasonality, the gap in worst ratios closes, but not because they forecast better: their smallest worst ratio is 1.58 (1.48 to 1.67, AutoARIMA), level with TimesFM-3’s 1.55 (1.41 to 1.71). AutoARIMA’s worst cell is still D4 at $n = 24$, and its mean MASE there is unchanged (2.30); what changes is the yardstick. At $n = 24$ the test cannot be applied, every series is treated as non-seasonal, and seasonal naive loses the true period that made it the best method in that cell (mean MASE 4.26 instead of 1.09), so TimesFM-3 (1.45) becomes the cell-best. Much of the classical deficit in worst ratios is thus measured against seasonal naive with a known period. With two cycles the automatic classical methods recover the seasonality in neither configuration: non-seasonal AutoETS has a mean MASE of 2.55 on D5 and 4.25 on D4, against 1.13 and 1.09 for seasonal naive with the true period and 1.11 and 1.45 for TimesFM-3.

A response-surface regression of the per-series $\log_2$ ratio of TimesFM-3’s MASE to AutoARIMA’s on the standardised parameters, $\log_2 n$ and the version, with standard errors clustered by parameter draw (Supporting Information, Tables S4 and S5, and Figure S2), shows that the dependence on the settings is modest. The largest systematic effect is length: each doubling of n moves the ratio against TimesFM-3 by 20% on D4 and 11% on D5, and in its favour on D6 and D8. Outliers favour TimesFM-3 on the seasonal processes (14% and 10%); the other version effects are at most 0.09 in absolute value. Among the parameters only the demand probability of D8 has a large effect; the ratio is flat in the autoregressive coefficient of D1 and the GARCH persistence of D9, grows in favour of TimesFM-3 with the size of the break in D6, and varies with

the inflection point of D7 in a way a linear term does not capture (Figure S2). The regressions explain little of the per-series variation ($R^2 \leq 0.26$).

7 | Application to M4 monthly data

Whether TimesFM-3 has seen the M4 data cannot be settled, but the documented corpus makes it less likely than for many public benchmarks: its real-world part is the GIFT-Eval pre-training collection (without the datasets overlapping fev-bench), Wikipedia page views and Google Trends [26], and that collection is built to exclude the GIFT-Eval test datasets, of which M4 Monthly is one [9, 63]. The exclusion works at the level of datasets, and the synthetic and augmented part of the corpus is not documented. TimesFM-2.5 lists the same sources [56]. The position of Chronos-Bolt differs: the original Chronos models were trained on M4 Monthly with its final 18 observations, the official test period, held out [5, Table 2]; if, as its model card suggests, Chronos-Bolt shares that corpus, its M4 forecasts are in-domain rather than zero-shot. The corpora of Chronos-2 and TiRex were not checked for M4, and their M4 results are read in the same way. Because no corpus can be inspected, the real-data tier is secondary evidence. This section reports a rolling-origin evaluation (Section 7.1), the official M4 test period (Section 7.2) and an experiment with truncated histories (Section 7.3); the last two are post hoc.

7.1 | Rolling-origin evaluation

A seeded sample of 1 000 series is drawn from the 32 581 M4 Monthly series with at least 138 training observations, enough for a context of 120 and the 18-month horizon. Their training periods have a median of 281 observations (interquartile range 198 to 306), against 202 (82 to 306) for all 48 000 series, so the sample over-represents the long end of the population. Each series is evaluated at up to three origins 12 months apart with M4’s horizon $h = 18$ (2 932 windows; the shortest series cannot support a third origin); consecutive windows overlap by six months, so the errors of a series are averaged over its origins and the series are the independent units of the paired Wilcoxon tests. TimesFM-3 and the pre-registered classical methods were run on this tier. TimesFM-3 (mean MASE 0.914) significantly outperforms seasonal naive (1.280; better on 80.1% of series), Theta (0.970; 56.8%) and AutoETS (0.951; 54.1%), and shows no detectable difference from AutoARIMA (0.923; 52.6%, adjusted $p = 0.27$) or the combination (0.907; 48.5%, adjusted $p = 0.27$). This ties TimesFM-3 with AutoARIMA on long, mostly seasonal series, rather than showing the 21% to 24% AutoARIMA advantage found on the simulated D4. The simulated seasonal process is a single, sharply seasonal model, whereas real series mix trend, seasonality and irregular components in proportions the simulation does not reproduce (Section 6.1); an advantage from contamination cannot be excluded either. The probabilistic results also diverge: TimesFM-3’s 80% intervals cover 0.765 against 0.801 for the combination, yet its pinball loss is the lowest of the six methods (0.363, against 0.365 for the combination and 0.374 for AutoARIMA). Its distribution is slightly better overall while its 80% interval is too narrow, the fragility of foundation-model calibration reported by Adler, Chang, Draxler, Abdi, and Smyth [11].

7.2 | The official M4 test period

Rolling origins cut from the training data cannot be placed against published M4 results, so the same series were also forecast for the official 18-month test period from their full histories and scored as in M4, with the overall weighted average (OWA) of sMAPE and MASE relative to the official Naive2 forecasts [3]. The comparison includes the methods of Section 5.2 and, post hoc, the published point forecasts of the ten best M4 submissions, among them the winning hybrid of exponential smoothing and a recurrent network [64] and the feature-based combination FFORMA [65]; scored on all 48 000 monthly series, the downloaded files reproduce the published M4 results.

Theta and the Comb benchmark improve on Naive2 by 7% to 9% in OWA (Table 8), in line with the competition. **Four M4 entries are ahead of every foundation model:** FFORMA (OWA 0.831), Jaganathan and Prakash (0.841), Smyl’s hybrid (0.848) and Pawlikowski et al. (0.849). TimesFM-3 (0.862; 0.857 with its evaluator settings) is level with the entries ranked fifth to seventh in M4 (0.858 to 0.864), TiRex (0.855) and Chronos-2 (0.860) are close to it, and the classical combination (0.868) and AutoARIMA (0.891) follow. On this sample the M4 entries’ OWA differs from their published monthly values by up to 0.03, as much as the gaps between them. A Friedman test on per-series ranks, with one configuration per model and without Naive2 [66], rejects the equality of the 22 methods, but ten lie within the Nemenyi critical difference of the best mean rank [60]: five M4 entries, the classical combination, TimesFM-3, Chronos-2 and TiRex. On real monthly series with long histories, zero-shot TimesFM-3 matches good specialised methods of 2018 without per-series tuning, but not the best of them.

TABLE 8 | M4 Monthly official test period: accuracy, OWA and multiple comparisons.

Method	sMAPE	MASE	OWA	Coverage	MSIS	Mean rank
FFORMA (M4, 2nd)	7.82	0.889	0.831	–	–	9.85*
Jaganathan & Prakash (M4, 4th)	7.88	0.902	0.841	–	–	10.06*
ES-RNN, Smyl (M4, 1st)	7.98	0.907	0.848	–	–	10.72*
Pawlikowski et al. (M4, 3rd)	7.95	0.912	0.849	–	–	10.38*
TiRex	8.04	0.916	0.855	0.801	4.40	10.77*
TimesFM-3, evaluator settings	8.01	0.923	0.857	0.765	4.50	–
Petropoulos & Svetunkov (M4, 6th)	8.09	0.916	0.858	–	–	10.46*
Chronos-2	8.11	0.917	0.860	0.783	4.47	10.69*
Fiorucci & Louzada (M4, 5th)	8.09	0.920	0.860	–	–	10.50*
TimesFM-3	8.05	0.929	0.862	0.761	4.54	10.69*
TimesFM-2.5	8.15	0.922	0.864	0.812	4.53	11.03
Shaub (M4, 7th)	8.06	0.933	0.864	–	–	11.01
Combination (ETS, ARIMA, DOTM)	8.16	0.928	0.867	0.789	4.74	–
Combination (Theta, ETS, ARIMA)	8.16	0.929	0.868	0.789	4.76	10.58*
TimesFM-2.5, settings off	8.25	0.932	0.874	0.802	4.61	–
Doornik et al. (M4, 9th)	8.21	0.945	0.878	–	–	11.64
Pedregal et al. (M4, 10th)	8.31	0.953	0.887	–	–	11.88
AutoARIMA	8.50	0.939	0.891	0.779	4.91	11.13
Chronos-Bolt	8.40	0.971	0.900	0.800	4.70	11.91
AutoETS	8.45	0.966	0.900	0.784	4.98	11.82
M4 Comb	8.50	0.982	0.911	–	–	12.65
Legaki & Koutsouri (M4, 8th)	8.51	0.995	0.917	–	–	12.76
DOTM	8.59	0.996	0.922	0.728	5.27	13.03
Theta	8.59	1.006	0.926	0.723	5.36	13.00
Naive2	9.22	1.092	1.000	–	–	–
Seasonal naive	10.39	1.306	1.161	0.780	6.56	16.44

Note: 1 000 M4 Monthly series, each forecast for the official 18-month test period from its full training history. sMAPE and MASE as defined in M4; OWA relative to the official Naive2 forecasts. Coverage and MSIS use the nominal 80% interval, the widest that every probabilistic method can form from nine deciles. Mean rank: average rank of per-series MASE among 22 methods, one configuration per model (Friedman $\chi^2 = 1054$, $p < 10^{-100}$); * marks methods whose mean rank is within the Nemenyi critical difference (1.04) of the best; – marks alternative configurations and Naive2, left out of the ranking. Published M4 submissions, ranked by their overall M4 position, provide point forecasts only [3, 64, 65].

7.3 | Truncated histories

The official test period was also forecast from only the last 24, 48 or 96 observations of each training series, with the full-history MASE denominator so that only the information available to the methods changes (Table 9). The simulation's estimability result reappears: with 24 observations, where the classical methods again have two cycles for a period of 12, the foundation models lead—TiRex 1.027, TimesFM-3 1.046, TimesFM-2.5 1.050—against 1.096 for the best classical method, the combination, and 1.175 for AutoARIMA; the difference was not tested. With 48 observations the combination is level with them (0.955 against 0.952 to 0.972), and with the full history the leading methods are within 2% of one another. Truncation also moves the information set away from the full series a model could have been trained on, although it cannot rule out that the series were seen.

8 | Operational considerations

Accuracy is not the only criterion. This section considers computational cost (Section 8.1) and licensing, interpretability and the basis of prediction intervals (Section 8.2).

8.1 | Computational cost

Timed separately on the same 200 seasonal series of length 96, seasonal naive took 3.1 ms per series, Theta 9.0 ms, TimesFM-3 23.2 ms (batched on an NVIDIA GTX 1660 Ti, excluding a one-off model load of 3.0 s), AutoETS 102.0 ms and AutoARIMA 2 382.7 ms, each classical method on one CPU core: automatic ARIMA searches over orders and fits each candidate by maximum likelihood, separately for every series, whereas TimesFM-3 amortises one forward pass over a batch. Per series, Theta and seasonal naive were thus faster than TimesFM-3 and AutoETS about four times slower, and

TABLE 9 | M4 Monthly with truncated contexts: mean MASE by context length.

Method	Last 24	Last 48	Last 96	Full
TimesFM-3	1.046	0.961	0.942	0.929
TimesFM-3, evaluator settings	1.045	0.960	0.936	0.923
TimesFM-2.5	1.050	0.972	0.952	0.922
Chronos-Bolt	1.158	1.023	0.965	0.971
Chronos-2	1.087	0.952	0.935	0.917
TiRex	1.027	0.956	0.949	0.916
AutoARIMA	1.175	1.019	0.979	0.939
AutoETS	1.129	0.996	0.995	0.966
Theta	1.150	1.011	1.003	1.006
DOTM	1.167	1.012	0.991	0.996
Combination (Theta, ETS, ARIMA)	1.096	0.955	0.947	0.929
Combination (ETS, ARIMA, DOTM)	1.111	0.959	0.945	0.928
Seasonal naive	1.306	1.306	1.306	1.306

Note: Mean MASE on the M4 official test period when each method sees only the last 24, 48 or 96 training observations, or the full history (median 281 observations). All columns use the full-history MASE denominator, so only the information given to the methods changes. Seasonal naive uses only the last 12 observations and is unaffected.

the classical methods parallelise trivially: in the main run, all four classical methods over the 7 200 series took 7.3 minutes on 22 cores, against 3.2 minutes for TimesFM-3 on one consumer GPU, a ratio of 2.3× that is dominated by AutoARIMA and depends on the hardware. Memory cannot be compared directly, because the profiler used sees neither the compiled kernels of the classical methods nor GPU memory; structurally, the classical methods hold a handful of parameters per series, whereas TimesFM-3 keeps its checkpoint of about 1.3 GB resident, whatever the number of series.

8.2 | Licensing, interpretability and prediction intervals

TimesFM-3’s weights are released under the TimesFM Non-Commercial License v1.0, which grants use “solely for your Non-Commercial Purposes” and requires a separate licence for any “commercial or production activity” [27, 26] (accessed 23 September 2026). The source code is Apache-2.0, as were the weights of TimesFM-2.5 and Chronos-Bolt, but the version-3 weights are not: **for a commercial application, TimesFM-3 is not licensed as released**, whatever its accuracy, and the alternatives are TimesFM-2.5, separate terms or a classical method. Licensing is routinely omitted from model comparisons although it decides which models an entire class of users may use. An ARIMA or ETS fit reports its orders, coefficients, smoothing parameters and standard errors, which support residual diagnostics, tests for remaining structure and an explanation of why a forecast turns where it does. TimesFM-3 exposes none of this: when it forecasts poorly, there is nothing to inspect. Its quantiles are learned outputs with no accompanying theory; they were well calibrated on average in the simulation (Section 4.5) but under-covered on M4, and their behaviour on a process unlike anything in the pre-training corpus is unknown. In regulated settings where a forecast must be justified, these differences may outweigh the accuracy comparison.

9 | Discussion

This section assesses the pre-registered hypotheses (Section 9.1), interprets the balance between robustness and peak accuracy (Section 9.2), gives practical recommendations (Section 9.3) and states the limitations (Section 9.4).

9.1 | Assessment of the pre-registered hypotheses

None of the five hypotheses survives unqualified: H1 and H5 are not supported, H4 is contradicted, and H2 and H3 are partially supported. The protocol did not state decision rules, so these verdicts are judgements on the pattern of pre-registered tests. **H1** (correctly specified classical models beat TimesFM-3 on their home ground, most at short lengths) is not supported as stated—at short lengths the gap runs the other way—and holds only where a model is both correctly specified and estimable (D4 and D5 at $n \geq 48$). **H2** (TimesFM-3 wins wherever no classical form is correct) is partially supported: mildly on breaks; not on the logistic process, where the outcome depends on length; and on intermittent

demand only against the classical methods, not against simple benchmarks built from the history. **H3** (on D9 the families tie in the mean and classical intervals under-cover) is partially supported: they tie, but only AutoARIMA under-covers. **H4** (the combination beats every classical method and is TimesFM-3's hardest opponent) is contradicted in both clauses: AutoARIMA beats the combination on mean MASE and is the harder opponent (11–9 against 27–3). **H5** (both families under-cover at 80%, TimesFM-3 more at longer horizons) is not supported: averaged over processes TimesFM-3 is close to nominal, its coverage does not fall with the horizon, and the classical methods over-cover from $n = 96$; per cell the deviations are large but not systematically downward.

9.2 | Robustness versus peak accuracy

The classical methods' 20 cell wins are spread over five methods, so which classical method wins is not knowable in advance without the model selection a foundation model is meant to remove. TimesFM-3, one untuned model, is not significantly worse than the best classical method in 25 of the 36 cells; the best classical method is ahead in the other 11, over five processes, by small margins on D2 and D3 and by large ones only on D4, D5 and D7 at $n = 96$ (these comparisons with a selected opponent are descriptive). The evidence therefore supports neither the conclusion that foundation models have superseded classical methods nor the opposite, but a narrower one: **in this design TimesFM-3 behaved robustly rather than with peak accuracy**. It is rarely the best method and never among the worst, whereas the automatic classical methods are sometimes excellent and sometimes catastrophic. For one well-understood series that trade is unattractive; for many heterogeneous series and no capacity to diagnose each one, it favoured TimesFM-3 here. The extended evidence qualifies the trade four times. Among the foundation models evaluated it belongs to TimesFM-3 alone (Section 5.2). It holds for short horizons only: at 48 steps AutoARIMA has the smaller worst case, and every foundation model mistakes a plateau for a decline (Section 5.5). Its size depends on the yardstick: much of the classical deficit is measured against seasonal naive with a known period, and without the two-cycle cells AutoARIMA's worst ratio is 1.62 against TimesFM-3's 1.31 (Section 4.3, Section 6.2). And on the M4 test period with long histories, TimesFM-3 is level with good but not the best M4 entries (Section 7.2).

9.3 | Practical recommendations

The recommendations below are conditional on the processes, lengths and horizon examined. They describe how the methods behaved here and are offered as diagnostic starting points to be checked on the user's own data, not as rules; several rest on considerations outside the accuracy comparison, and where the evidence does not favour one family, more than one option is named.

Many heterogeneous series without per-series tuning. This is where TimesFM-3's robustness mattered most in this design: it won most significant comparisons against every classical method except AutoARIMA and was never far from the best method of a cell. Where a model can be identified for a single series, the classical route loses little—against AutoARIMA the comparisons split 11–9—and keeps parameters, diagnostics and interval theory that can be explained and audited.

Seasonal series. With at least four observed cycles, AutoARIMA was 21–24% more accurate than TimesFM-3 on the seasonal ARIMA process D4 and 4–12% on the ETS process D5, and seasonal naive was also ahead on D4. With two cycles the automatic classical methods fell back to non-seasonal models and failed, and seasonal naive, or TimesFM-3, was the safer choice.

Intermittent demand. Simple benchmarks built from the history were as good as anything tested: the empirical deciles of the history gave the lowest pinball loss, and its mean, like the Croston family, the lowest RMSSE. TimesFM-3 came close on both only with a suitable read-out of its deciles, and its median forecast was no better than forecasting zero. These results come from a process that is independent over time, the case most favourable to history-based benchmarks.

Long horizons. Beyond about a year of monthly steps the evidence turns: AutoARIMA had the smaller worst case, and every foundation model forecast a decline on a process that had levelled off, so long-horizon forecasts from these models should be checked against the recent level before use.

Intervals, cost and licence. TimesFM-3 had the best 80% interval score in 22 of the 36 cells, including every length after a structural break (D6); on M4, however, its 80% intervals under-covered (0.765 against 0.801 for the classical combination on the rolling origins), so interval quality on real data should be checked rather than assumed. On the hardware used, TimesFM-3 on a GPU was faster end to end than the parallelised classical suite, a comparison dominated by AutoARIMA. For commercial deployment the licence decides before accuracy does: TimesFM-3's weights are non-commercial, and the Apache-2.0 TimesFM-2.5 was less robust in the simulation (worst ratio 2.18) though level with TimesFM-3 on M4.

526 9.4 | Limitations

527 First, simulated series are clean and free of the measurement error, calendar effects and regime changes of applied work;
528 the M4 tier addresses this only in part, and the randomised design still leaves about a quarter of M4 monthly series, and all
529 other frequencies, outside the regions studied, with coverage in four features a necessary rather than sufficient condition
530 for similarity. Second, the nine process families are fixed, the pre-registered horizon is short ($H = 12$) and the longer one
531 was examined in one secondary check, the seasonal period is 12, and the departures from the assumptions were applied
532 one at a time, so their interactions were not examined. Third, TimesFM-3 was used zero-shot and univariate, leaving its
533 multivariate and covariate capabilities untested; fine-tuning was excluded because fine-tuning on one series of 24 to 200
534 points is not meaningful for a model of this size and fine-tuning on the DGPs would remove the separation between model
535 and test data. Fourth, no method was tuned by hand, and the classical methods were run in StatsForecast at its defaults;
536 a cross-check with forecast on the seasonal processes reproduced the failures but found one implementation failure, and
537 an expert who identified each process could select better models. Fifth, five foundation models were evaluated and they
538 behave differently, so conclusions about one do not transfer to the class; masked-encoder models such as Moirai [6] were
539 not included. Sixth, the summaries that carry the robustness finding—worst ratios and win counts—were defined after
540 the protocol and depend on the methods and cells compared. Seventh, the protocol was frozen before any forecast but held
541 in the replication archive rather than deposited with an external registry, and several analyses are post hoc—the Croston
542 and history benchmarks, the combination rule, the Oracle ARIMA, Sections 5 and 6, and the official-test and truncation
543 analyses of Section 7—and are reported separately from the pre-registered comparison.

544 10 | Conclusion

545 This study compared TimesFM-3 with classical methods on 7 200 series from nine known processes under a pre-registered
546 protocol, in a design where realisation-level contamination is impossible and the model families searched by AutoARIMA
547 and AutoETS contain the true process of D1–D5. Neither family dominates. TimesFM-3 wins 113 of 132 significant
548 pairwise comparisons, only AutoARIMA holding its own (11–9), and it is never more than 1.31 times worse than the
549 best method in a cell (95% interval 1.28 to 1.36), while every classical method is at least 2.2 times worse somewhere—
550 mostly with two seasonal cycles, where the automatic methods fall back to non-seasonal models; without those cells
551 AutoARIMA's worst ratio is 1.62. With at least four seasonal cycles AutoARIMA is 21–24% more accurate. Among the
552 five foundation models evaluated this robustness is specific to TimesFM-3, randomised parameters, heavy tails and out-
553 liers leave it qualitatively unchanged, and it holds for short horizons only: at 48 steps AutoARIMA has the smaller worst
554 case and every foundation model forecasts a decline on a saturated process. On intermittent demand TimesFM-3 shows
555 no advantage over simple benchmarks built from the history.

556 On the official test period of 1 000 M4 monthly series TimesFM-3 (OWA 0.862) is level with the M4 entries ranked fifth to
557 seventh and behind the four best, and with only 24 observations of history its mean MASE is 4.6% below that of the best
558 classical method. The finding most likely to transfer concerns the classical side, where the mechanism can be examined:
559 what matters is whether a classical model is *estimable* from the data, not only whether it is correctly *specified*—with two
560 cycles AutoETS almost never selected a seasonal form and had 2.7 times the error of seasonal naive on data from an ETS
561 process. Outside accuracy, TimesFM-3 was 2.3 times faster end to end than the parallelised classical suite on the hardware
562 used, and its weights are licensed for non-commercial use only, which for commercial users decides the question before
563 accuracy does.

564 The study is exploratory. It describes how one black-box model behaves on data of known structure, does not explain that
565 behaviour, and remains conditional on the processes, parameter ranges, horizon and benchmarks examined. Studies of
566 other foundation models, frequencies and multivariate problems, and of the internal representations of these models, will
567 be needed before the boundary between learned and parametric forecasting can be stated in general terms.

568 11 | Computational details and reproducibility

569 All results were produced with Python 3.13 [67], statsforecast 2.1.1 [38], timesfm 3.0.1 with the google/timesfm-3.0-p
570 ytorch weights (revision 43046b8) [27], chronos-forecasting 2.3.2 with amazon/chronos-bolt-base (revision 5d9f166)
571 and amazon/chronos-2 (revision 29ec376), tirex-ts 1.4.2 with NX-AI/TiRex (revision 63c7409), the google/times
572 fm-2.5-200m-pytorch weights (revision 1d95242), R 4.4.1 with forecast 9.0.2 for the cross-check, PyTorch 2.6.0 with
573 CUDA 12.4 and cuDNN 9.1 [68], NumPy [69], pandas [70], SciPy [71] and Matplotlib [72], on an NVIDIA GTX 1660 Ti
574 (driver 591.86) with a 24-core CPU and 32 GB of memory; parallel classical runs use 22 worker processes. The full package

list is pinned in the replication archive. Every simulated series is seeded from its own identifiers, so the study regenerates exactly, independently of core count or execution order, and each cached stage can be rerun alone. The foundation models were run in 32-bit arithmetic without random sampling; rerunning two TimesFM-3 cells reproduced the stored quantiles exactly, although other hardware may differ in the last digits because GPU floating-point reductions are not associative. The protocol, with its five hypotheses, was frozen on 8 September 2026, before any forecast was produced; it is in the replication archive with a dated log of the fourteen departures from it, each reasoned, and every post-hoc analysis is marked as such. Every DOI in the bibliography was resolved and compared with its entry.

ACKNOWLEDGEMENTS

First and foremost, the authors thank Allah, the Almighty, for everything He has given them, including the ability to complete this work. The authors used a large language model to improve the language and readability of this manuscript and to assist with the implementation of the simulation code; they reviewed and edited all content and take full responsibility for it.

FUNDING

No funding was received for this work.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

ETHICS APPROVAL

Not applicable; the study uses simulated data and the publicly available M4 competition data, and involves no human participants. Permission to reproduce material from other sources: not applicable.

DATA AVAILABILITY STATEMENT

The complete research compendium is archived on Zenodo at 10.5281/zenodo.22681072 (concept DOI, resolving to all versions): the pre-registered protocol, the deviation log, the entire pipeline, the per-series metrics for all 129 600 evaluations, the raw forecasts and quantiles, every figure and the manuscript source. All data in the simulation tier are generated by that code from documented seeds; the real-data tier uses the publicly available M4 competition data [3], which the code re-fetches, with the seeded 1 000-series sample included so the analysis reproduces exactly. The raw forecasts are deposited deliberately and not only for completeness. This paper’s intermittent-demand result reverses between absolute- and squared-error metrics, so a reader who wants to re-score the same forecasts under a different loss can do so without re-running the study. TimesFM-3’s weights are not redistributed: they are released by Google under a non-commercial licence and are downloaded by the code from Hugging Face.

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FIGURE LEGENDS

- Figure 1.** One realisation of each data-generating process at $n = 96$. The context supplied to every method is in grey and the held-out horizon in colour, separated by the dotted line.
- Figure 2.** Accuracy of TimesFM-3 relative to the best classical method in each cell, on a $\log_2$ scale. Bars below zero mark cells where TimesFM-3 is more accurate. The label under each process names the family correctly specified for it by construction.
- Figure 3.** Pairwise comparisons of TimesFM-3 with each classical method, cell by cell: median of the paired per-series ratios of MASE over $h = 1, \dots, 12$, with 95% bootstrap intervals, on a logarithmic scale. Filled markers are significant after Benjamini–Hochberg correction; points left of 1 favour TimesFM-3. Arrowheads mark medians beyond the plotted range.
- Figure 4.** Empirical coverage of the nominal 60% and 80% intervals, averaged over all nine processes, against series length. The nominal level is dashed. TimesFM-3’s 80% coverage stays within 0.802–0.834 across every length; the classical methods under-cover at $n = 24$ and over-cover from $n = 96$.
- Figure 5.** Simulated and M4 Monthly series in a four-feature instance space (spectral entropy, trend strength, seasonal strength and first-order autocorrelation of the differences), first two principal components. Grey points are M4 series; coloured points are simulated series at $n = 96$. (a) The fixed-parameter design covers 48.5% of the M4 series; (b) the randomised-parameter design covers 75.8%, with coverage defined in Section 6.1.